

CBSE Practice Worksheet

Class 8 | Science | Difficulty: Hard | Set A

Instructions: Answer all questions. This worksheet contains 25 questions including multiple choice, fill in the blanks, and short answer questions.

Section A: Multiple Choice Questions (1 mark each)

1. Which organelle is involved in programmed cell death?
(A) Lysosome
(B) Mitochondria
(C) Peroxisome
(D) Ribosome
2. What is the wavelength of Lyman series limit?
(A) 91.2 nm
(B) 121.6 nm
(C) 656.3 nm
(D) 364.6 nm
3. What is the hybridization of carbon in methane?
(A) sp
(B) sp^2
(C) sp^3
(D) sp^3d
4. What is the coordination number in FCC structure?
(A) 6
(B) 8
(C) 12
(D) 14
5. Which law relates induced EMF to rate of change of magnetic flux?
(A) Lenz's law
(B) Faraday's law
(C) Ampere's law
(D) Ohm's law
6. What is the efficiency of an ideal Carnot engine operating between 300K and 400K?
(A) 25%
(B) 33%
(C) 50%
(D) 75%
7. What is the wavelength of de Broglie wave for an electron?
(A) $\lambda = h/mv$
(B) $\lambda = mv/h$
(C) $\lambda = hv/m$
(D) $\lambda = m/hv$
8. Which reaction is endothermic?
(A) Combustion
(B) Neutralization
(C) Photosynthesis

(D) Respiration

9. What is the oxidation state of Mn in KMnO_4 ?

- (A) +2
- (B) +4
- (C) +6
- (D) +7

10. Which principle explains the working of a hydraulic lift?

- (A) Archimedes
- (B) Pascal
- (C) Bernoulli
- (D) Newton

Section B: Fill in the Blanks (1 mark each)

11. In $\text{S}_\text{N}2$ reactions, the rate depends on both ____ and nucleophile.

12. Raoult's law applies to ____ solutions.

13. The Krebs cycle occurs in the ____ of mitochondria.

14. The magnetic field inside a solenoid is given by $B = \mu \blacksquare$ ____.

15. The process of splitting water during photosynthesis is called ____.

16. The enzyme that converts prothrombin to thrombin is ____.

17. For a reversible process, entropy change of the universe is ____.

18. The IUPAC name of $\text{CH}_3\text{CH}_2\text{CHO}$ is ____.

Section C: Short Answer Questions (2 marks each)

19. Derive the lens formula and explain its applications.

20. Describe the mechanism of nerve impulse transmission.

21. Explain hybridization with examples of sp , sp^2 , and sp^3 .

22. Derive the equation for kinetic energy of gases from kinetic theory.

23. Explain Le Chatelier's principle with examples.

24. Describe the light-dependent and light-independent reactions of photosynthesis.

25. Explain the mechanism of DNA replication.

--- End of Worksheet ---

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